TRIBHUVAN UNIVERSITY

FACULTY OF HUMANITIES AND SOCIAL SCIENCES

An Internship Report On

Frontend Development

At
Code Bridge (Everest Adventure Nepal)

In partial fulfillment of the requirements of Bachelor of Computer Application

Under the Supervision of

Mr. Bijay Shrestha

Submitted by:

Aaditya Shrestha

Symbol No:

101402082

Registration No:
6-2-1014-67-2021

Submitted to:

Tribhuvan University

Faculty of Humanities and Social Sciences

March, 2026

SUPERVISOR'S RECOMMENDATION

I hereby recommend this internship work prepared under my supervision by **Aditya Shrestha** in partial fulfillment of the requirements for the degree of Bachelor of Computer Application for the final evaluation. I have thoroughly looked over the work he has done during his internship period.

____

Mr. Bijay Shrestha

SUPERVISOR

Department of Information

Technology,

NIST College Banepa-9, Nayabasti, Nala,

Kavrepalanchok

MENTORS'S RECOMMENDATION

PAN NO. 13 127 7492

Ref. No.: 011 **Date:** 24th Mar, 2026

TO WHOM IT MAY CONCERN

This is to certify that **Mr. Aaditya Shrestha** has successfully completed a 3-month internship as a Next JS Intern at Code Bridge from 18th November, 2025 to 27th February, 2026.

During the internship period, the intern was actively involved in developing dynamic and responsive user interfaces using Next JS, building reusable components, and integrating APIs to create efficient and user-friendly web applications. The intern also gained hands-on experience with modern frontend development practices, including component-based architecture, state management, hooks, and UI/UX design principles.

Throughout the internship, **Mr. Aaditya Shrestha** demonstrated dedication, strong learning ability, and a positive attitude towards work. The intern was able to complete assigned tasks effectively and collaborated well within the development team.

We appreciate the efforts and contributions made during the internship and wish him all the best for his future academic and professional endeavors.

Bijay Shrestha
Founder & CEO
Code Bridge

984-9689512 code.bridge.team@gmail.com Banepa-8

LETTER OF APPROVAL

We recommend the internship work entitled "Everest Adventure Nepal**"** submitted by **Mr Aditya Shrestha (101402082)** in partial fulfillment of the requirements of Four Years Bachelor Degree of Computer Application has been examined by us and accepted for the award of degree under Tribhuvan University.

______________________ **SIGNATURE of Supervisor** Mr. Bijay Shrestha Department of Humanities and Social Science Nist College Banepa, Kavre	______________________ **SIGNATURE of HOD/ Coordinator** Mr. Samish Shrestha Department of Humanities and Social Science Nist College Banepa, Kavre
______________________ **SIGNATURE of Internal Examiner** Internal Examiner	______________________ **SIGNATURE of External Examiner** External Examiner

ACKNOWLEDGEMENT

This intern report is prepared in partial fulfillment of the requirements for the degree of Bachelor of Computer Application (BCA). The satisfaction and success of completing this task would be incomplete without heartfelt thanks to the people whose constant guidance, support, and encouragement made this work successful.

Firstly, I would like to express my gratitude towards my mentor **Mr.Bijay Shrestha,** a backend developer at **Code Bridge** for his unquestionable support. Without his support and encouragement, it would have been difficult to work on.

I would also like to thank my supervisor **Mr.Bijay Shrestha**, lecturer of **NIST College, Banepa** for his invaluable encouragement, guidance, and ever willingness to spare time from his otherwise busy schedule.

I would like to dedicate our hearty gratitude to **Code Bridge,** for providing us with an opportunity to do an internship at this reputed organization with full support and cooperation.

In the end, I would like to express my sincere thanks and appreciation to all my colleagues and seniors who have helped me directly or indirectly during this internship period. I would like to make them part of my success.

Sincerely,

Aaditya Shrestha

(TU Reg no.:6-2-1014-67-2021)

ABSTRACT

This internship report presents the development of the *Everest Adventure Nepal* web application carried out at Code Bridge as partial fulfillment of the requirements for the Bachelor of Computer Application (BCA) degree. The internship was conducted for a duration of three months, focusing on Frontend Development and API implementation. The *Everest Adventure Nepal* platform is designed to facilitate the booking of Everest trekking packages, including luxury and semi-luxury options, providing users with a complete 14-day trekking tour experience. The development process followed the Agile Software Development Life Cycle (SDLC) methodology to ensure flexibility, continuous improvement, and efficient delivery of the application. Various tools and technologies were used during the development process, including Next.js, Tailwind CSS, and Shadcn UI for frontend development. The internship involved designing and developing user interfaces, performing data validation, integrating APIs, and implementing Stripe payment functionality. The internship was completed under the supervision of Bijay Shrestha, providing valuable exposure to real-world software development practices, team collaboration, and professional standards in the IT industry.

Keywords: Agile SDLC, Frontend Development, API Integration, VS Code, UI/UX Design, Figma, Web Application, IT

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LIST OF ABBREVIATIONS

API	Application Programming Interface
BCA	Bachelor Of Computer Application
ICT	Information and Communication Technology
IDE	Integrated Development Environment
IT	Information Technology
SDK	Software Development Kit
SDLC	Software Development Life Cycle
UI	User Interface

CHAPTER 1: INTRODUCTION

1.1 Introduction

The internship program is designed to provide students with the opportunity to gain real-world industry experience while working toward achieving their professional goals and career aspirations. The primary objective of an internship is to enable students, referred to as interns, to apply theoretical knowledge acquired during their academic studies to practical situations within a professional working environment.

The Bachelor of Computer Application (BCA) program requires final-year students to complete a six-credit internship with a minimum duration of eight weeks or 300 working hours as a mandatory academic requirement. This internship may be completed in either government or private sector organizations. Furthermore, feedback collected from host organizations plays a crucial role in evaluating student performance and enhancing the overall quality and professionalism of BCA graduates.

During the internship period, the intern was assigned to the software development department of Everest Adventure Nepal under the supervision of the assigned supervisor. The primary project on which the intern contributed was the Everest Adventure Nepal web-based trekking booking platform.

Everest Adventure Nepal is an Australia-based online trekking booking company that provides a digital platform for booking Everest Base Camp trekking packages, including luxury and semi-luxury 14-day trekking tours. The platform connects trekkers directly with experienced local guides and porters, ensuring transparency, fair payment, and ethical tourism practices. The core objective of the platform is to make Everest Base Camp trekking accessible, affordable, and trustworthy, without hidden costs.

The application is designed to provide users with a seamless and user-friendly experience for browsing trekking packages, booking tours, and accessing detailed trekking information from any location at any time. Through this project, the intern gained practical exposure to real-world web application development, backend system design, and professional software development practices.

The mission of Everest Adventure Nepal is to deliver safe, memorable, and authentic adventure experiences, while supporting and empowering local communities, promoting responsible and eco-friendly tourism, and preserving Nepal’s natural and cultural heritage. The company envisions becoming the most trusted and globally recognized adventure travel company in Nepal, leading sustainable mountain tourism practices, inspiring a global community of mountain enthusiasts, creating positive economic impact in rural Nepal, and setting new standards of excellence in adventure tourism.

1.2 Problem Statement

Planning and booking an Everest Base Camp trekking experience is often complex and time-consuming due to lack of transparent pricing, unreliable intermediaries, hidden costs, and limited access to verified guides and porters. The absence of a centralized and user-friendly online platform further makes it difficult for users to compare trekking packages and complete secure bookings. To overcome these challenges, the *Everest Adventure Nepal* platform provides a reliable, transparent, and technology-driven web application that simplifies trekking bookings, ensures fair compensation for local workers, and promotes sustainable tourism.

1.3 Objective

This internship program was done to fulfill the academic requirement of the BCA 7th Semester. An internship provides a variety of benefits for young workers who want to broaden their chances of landing a job and jump-starting their careers. The main objective of this internship project is to gain practical work experience in the field of backend development.

1.3.1 Internship Program Objective

The following is the list of objectives that the internship may fulfill:

- To develop an ability to work in a team.
- To gain practical work experience in the field of backend development.
- To enhance the intern's professional readiness to work in the real-world IT industry.

1.3.2 Internship Project Objective

The broad objectives of this project are as follows:

1. To develop a user-friendly web platform for booking Everest Base Camp trekking packages.
2. To ensure transparent pricing and promote sustainable, ethical tourism through digital solution

1.4 Scope and Limitations

1.4.1 Scope

An internship provides an opportunity to develop skills and gain practical experience in a specific field, helping individuals explore different roles and identify their long-term career interests. It enhances a resume, offers exposure to the professional working environment, and helps build valuable networks that support career growth. Internships also enable participants to gain a deeper understanding of their chosen field.

During my internship, I had the opportunity to contribute to the frontend development of a web application named *"Everest Adventure Nepal."* The frontend system focused on designing and implementing user-friendly interfaces for core functionalities such as user interaction, trekking package browsing, and booking processes. Modern frontend technologies were used to create responsive and visually appealing web pages. The work involved developing interactive user interfaces, performing client-side data validation, integrating APIs to connect with the backend system. This project provided practical exposure to frontend development, UI/UX design principles, API integration, and real-world web application development in a professional environment.

1.4.2 Limitations

Some drawbacks the intern faced during the internship are:

- It was challenging to balance internship work and college at the same time.
- Economic details of the project have not been mentioned due to confidentiality issues.
- Not all facts of the organization's operation can be disclosed due to restrictions imposed by the privacy regulations of the organization.

1.5 Report Organization

This report has been organized into 5 chapters:

Chapter 1: Here, an Introduction about the internship and my project with the statement of the problem has been stated. The scope and limitations of the internship are listed along with the objectives of the internship and what it hopes to achieve.

Chapter 2: Here, I have introduced the company details including the organization's structure, and its area of operation. Also, this chapter contains the descriptions of related works that I enrolled.

Chapter 3: Here, I have illustrated the previous work related to the project, and similar works were studied and different feasibility analyses are summarized in this section.

Chapter 4: Here, I have illustrated the roles and responsibilities I had at the organizations. Things such as the Weekly log, Description of the Project Involved during the Internship, Tasks/Activities, and requirements have been clarified.

Chapter 5: This section concludes the project that I have enrolled in. It gives a summary of what system was developed and the learning outcomes of the internship including references. The lesson learned is also included in this chapter.

CHAPTER 2: INTRODUCTION TO ORGANIZATION

2.1 Introduction to Organization

Code Bridge Software Company & IT Institute was established in 2018 in Nepal with the objective of providing professional software development services and quality IT education. The organization offers a wide range of services including custom software development, mobile and web application development, IT training programs, technical consultation, and digital solutions tailored to the specific needs of individuals and organizations.

Code Bridge focuses on understanding real-world problems and transforming them into effective, user-friendly, and scalable software solutions. The company emphasizes innovation, practicality, and quality while delivering customized applications that support business growth and digital transformation. In addition to software development, Code Bridge plays a significant role in skill development by offering industry-oriented training programs that bridge the gap between academic learning and professional practice.

The organization is supported by a team of skilled software engineers, instructors, and technical professionals who are committed to delivering high-quality solutions and services. Code Bridge is dedicated to adding value for its clients and learners by providing reliable software products, technical expertise, and continuous support, helping them achieve their technological and business objectives.

Table 1: Overview of Company Profile

Organization Name	Code Bridge
Address	Banepa_8
Contact	9849689512
Year of Establishment	2018
Email	Code.bridge.team@gmail.com
Website	https://codebridge.com.np/

Figure 1.1: Logo of Organization

2.2 Organization Hierarchy

Code Bridge comprises an administrative team along with interns, junior and senior programmers, and web designers.

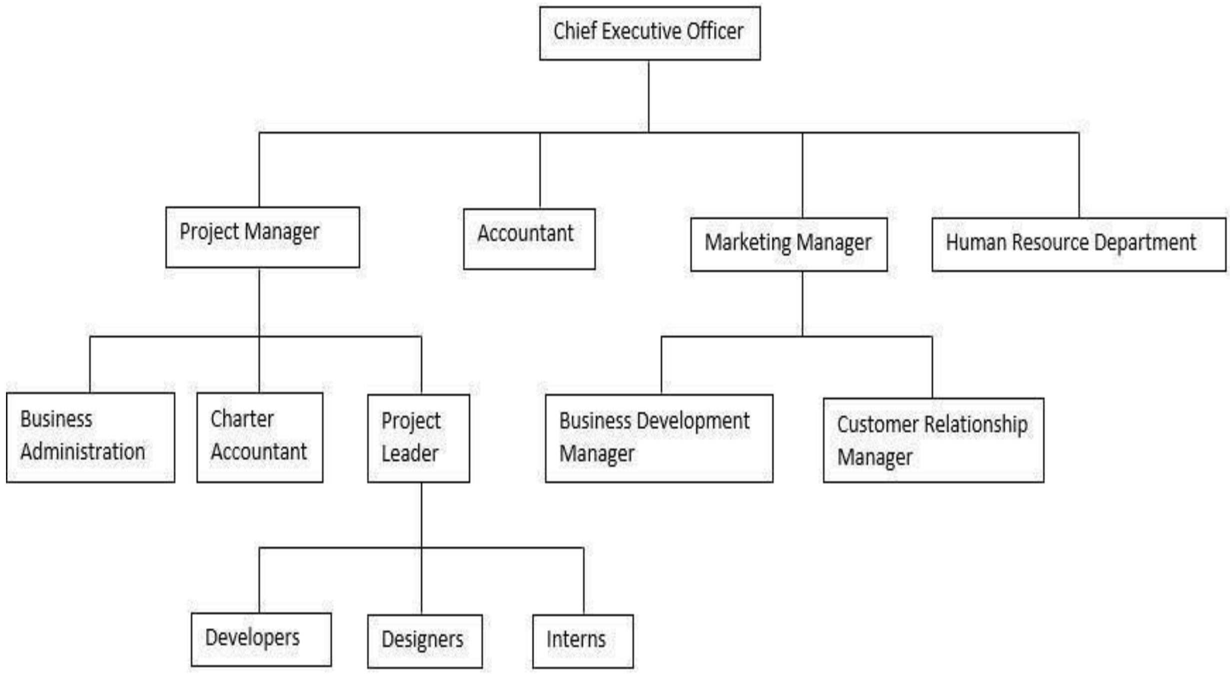

Figure 2.2: Organizational Hierarchy

2.3 Working Domains of Organization

Code Bridge is a software development company delivering services at the forefront of the evolving practices in software development. The management team at Code Bridge is strong, and the organization is supported by skilled software engineers and IT professionals with experience across various ICT domains. These include software development, business application solutions, and digital marketing–related technologies. The company’s professionals possess the technical expertise and practical knowledge required to deliver reliable and efficient digital solutions. Code Bridge emphasizes innovation, quality, and client satisfaction while addressing real-world technological challenges through modern software development practices.

- Web Development
- Mobile Application Development
- Portfolio site development
- M-Commerce Solutions

- Finance Management Applications
- Booking and ticketing apps
- UI/UX
- Healthcare management software
- Billing software
- Software as per customer requirements

2.4 Description of Intern Department

2.4.1 Introduction

The intern was placed in the Development department. It deals with researching, ideating, and designing different mockups, layouts, and designs for various products and developing the system according to the client's requirements and specifications. All in all, the department deals with all the research, designing & development-related works of the product in the organization.

2.4.2 Mentors

After joining the Developers Team at Code Bridge, the intern got an opportunity to work and be supervised by a professional mentor, **Mr. Bijay Shrestha** who has been working in this field for many years and helped the intern and guided them through their internship period.

2.4.3 Team

In Code Bridge, the intern got an opportunity to work in a team for the first time in their professional career. The intern was guided by a very professional mentor and got the opportunity to work under the expertise of the mentor and a supportive team which helped them to develop and nourish their skills and experience new things.

Table 2: Internship Period Details

Start Date	18th November,2025
End Date	27th February, 2026
Position	Java Developer Intern
Working Hours	7 hours a day
Office Days	Sunday- Friday

CHAPTER 3: BACKGROUND STUDY AND LITERATURE REVIEW

3.1 Background Study

An internship is a period of practical work experience offered by an organization to students, usually undergraduates, to provide exposure to a real working environment related to their field of study. Internships help students gain practical skills, industry knowledge, and professional experience while allowing organizations to benefit from the intern`s contributions. Internships may vary in duration and may be paid or unpaid depending on the organization. With a strong emphasis on learning and training, internships provide students with real-world exposure, help them develop essential workplace skills, and assist them in making informed career decisions. They also enable students to build professional networks and serve as an important stepping stone toward future employment.

I completed a three-month internship at Code Bridge, where I gained valuable hands-on experience in Next.js frontend development. During this internship, I developed a strong understanding of frontend application development using Next.js. I worked on designing and developing user interfaces, integrating RESTful APIs, handling client-side logic, and supporting core frontend functionalities of web applications. In addition to technical skills, the internship helped me improve my time management, sense of responsibility, teamwork, and professional communication skills. This experience significantly enhanced my understanding of industry practices and strengthened my foundation in Next.js Frontend software development.

3.2 Literature Review

The success of any web-based application is closely related to its usability, performance, and responsiveness from the user's perspective. A well-designed frontend plays a crucial role in ensuring an intuitive user interface, smooth navigation, and seamless interaction between users and the system. Research highlights that applications with user-centered and responsive frontend designs are more likely to achieve higher user satisfaction, better engagement, and improved overall performance (Bass et al., 2013). Therefore, focusing on effective frontend development is essential for the success of an online trekking booking platform.

The rapid growth of digital tourism and online booking platforms has significantly transformed how travellers plan and book trekking experiences. With increased access to the internet and digital services, users prefer platforms that offer clean interfaces, easy navigation, and interactive features for trekking bookings due to convenience and accessibility (Gretzel et al., 2015). Frontend technologies enable users to browse trekking packages, view itineraries, compare options, and complete bookings efficiently, thereby enhancing the overall travel planning experience.

As trekking booking platforms handle sensitive information such as personal details and payment data, ensuring secure and user-friendly frontend interactions is critical. Studies emphasize the importance of secure authentication interfaces, proper form validation, and safe handling of user inputs to enhance trust and usability (Kaur & Singh, 2020). This project acknowledges the need for implementing proper validation techniques and secure integration with APIs and payment systems to ensure a reliable user experience.

Overall, this project aims to apply established research findings and industry best practices to develop a responsive, user-friendly, and efficient frontend system for the Everest Adventure Nepal web application. By emphasizing UI/UX design, frontend performance, API integration, and user interaction, the application seeks to improve the online trekking booking experience and support the growing demand for digital adventure tourism platforms.

Existing Systems in Online Trekking and Travel Booking in Nepal

Several online trekking and travel booking platforms are currently operational in Nepal, offering digital services for trekking and adventure tourism. These platforms aim to simplify trip planning, package selection, and booking processes for domestic and international trekkers.

Bookmundi is an international travel booking platform that provides trekking and tour packages in Nepal, including Everest Base Camp treks. It allows users to browse itineraries, compare packages, and book trips online, but often involves third-party operators, which may reduce pricing transparency (Bookmundi, 2023).

Trekking Agencies' Association of Nepal (TAAN)–affiliated agencies operate individual websites to promote trekking packages and services. While these platforms provide detailed trekking information, many lack centralized booking systems, real-time availability, and transparent cost breakdowns (TAAN, 2023).

Viator and GetYourGuide are global travel platforms that offer trekking and adventure experiences in Nepal. These platforms benefit from strong international visibility and secure payment systems but usually charge higher service fees and rely on intermediaries, which may limit direct communication with local guides (Viator, 2023).

Local trekking agency websites in Nepal provide direct booking options and customized trekking services. However, many of these platforms have limited technical features, lack standardized pricing, and do not always ensure fair payment mechanisms for guides and porters.

Compared to existing platforms, Everest Adventure Nepal focuses on providing a centralized, transparent, and technology-driven trekking booking system that connects trekkers directly with verified local guides and porters. The platform emphasizes ethical tourism, fair compensation, and a user-friendly, secure frontend experience, ensuring intuitive navigation, responsive design, and seamless interaction with backend services.

CHAPTER 4: INTERNSHIP ACTIVITIES

4.1 Roles and Responsibilities

During my internship period, I was entrusted with a range of responsibilities by my mentor, Mr. Bijay Shrestha. These responsibilities encompassed various tasks vital to the internship experience. Initially, my mentor assigned me the task of studying and conducting research on the forthcoming application that was to be developed. This phase allowed me to acquire a firm grasp of the foundational aspects of app development. Subsequently, I was responsible for implementing the core functionalities that needed to be integrated into the project.

As a Frontend Developer (Intern) at Code Bridge, the following tasks and responsibilities were assigned during the internship period:

- Understand the organizational workflow, frontend development practices, and UI/UX design standards.
- Complete frontend-related tasks assigned by the supervisor on a weekly basis.
- Provide regular progress updates and reports to the supervisor.
- Interpret UI/UX designs and prototypes created in Figma and implement them accurately in Next.js.
- Develop and maintain interactive user interfaces using Next.js.
- Implement responsive layouts and design elements using Tailwind CSS and Shadcn UI.
- Integrate frontend components with backend APIs to ensure seamless data flow.
- Identify, handle, and debug errors in frontend logic and API responses.
- Perform client-side validation and ensure optimal performance across devices.
- Write and execute frontend test cases to ensure UI reliability and functionality.
- Document frontend components, interface designs, and implementation details for better collaboration.
- Follow professional discipline, teamwork, and ethical workplace behavior.
- Practice effective use of development tools, technologies, and version control systems (e.g., Git, VS Code).
- Build professional relationships and maintain effective communication within the development team.

4.2 Weekly log

During my internship period at Code Bridge, I was engaged in hands-on learning and practical work focused on Next.js frontend development. The internship followed a six-day workweek from 10:00 AM to 5:00 PM, which allowed me to gain in-depth experience in UI/UX implementation, responsive interface design, API integration, and frontend performance optimization. The following table summarizes the weekly log of activities carried out during the internship period:

Table 3: Weekly Log

Week 1

Date	Activities
Nov 18 to Nov 24	• Orientation and introduction to the organization • Setup of development environment (Node.js, VS Code, Git) • Introduction to frontend development concepts • Revision of HTML, CSS, and JavaScript fundamentals

Week 2:

Date	Activities
Nov 25 to Dec 1	• Learned modern JavaScript features • Introduction to React and Next.js basics • Setup Next.js project and routing

Week 3:

Date	Activities
Dec 2 to Dec 8	• Studied Next.js architecture(pages,components) • Created layout and navigation components • Implemented static and dynamic components

Week 4:

Date	Activities
Dec 9 to Dec 15	• Learned Tailwind CSS utility classes • Created responsive layout using Tailwind • Built reusable UI components with Shadcn UI

Week 5:

Date	Activities
Dec 16 to Dec 22	Started real project: **Everest Adventure Nepal** • Implemented homepage and navigation design • Built trekking package listing components • Applies design consistency with Tailwind & Shadcn UI

Week 6:

Date	Activities
Dec 23 to Dec 29	• Worked with Figma designs and translated them to code • Create responsive grids and card layout • Managed client side state using React hooks

Week 7:

Date	Activities
Dec 30 to Jan 5	• Integrated frontend with backend REST APIs • Used postman to inspect API responses for integration • Enhanced error handling and loading states

Week 8:

Date	Activities
Jan 6 to Jan 12	• Built Authentication UI • Implemented secure form validation • Managed protected routes and session handling

Week 9:

Date	Activities
Jan 13 to Jan 19	Final Integration and Testing • Performed UI testing and debugging • Ensured responsive across devices • Reviewed and optimized UI codes

Week 10:

Date	Activities
Jan 20 to Jan 26	• Finalized remaining UI modules • Built “Get in Touch” and Testimonals UI components • Prepared final deliverables and documentation

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Supervisor’s Signature

Mr. Bijay Shrestha

4.3 Description of the Project

During my internship, I had the opportunity to participate in several projects, among which the most significant was the development of the *Everest Adventure Nepal* web application. In the initial phase, I collaborated with my mentor and the development team to understand the system requirements, frontend workflow, UI/UX designs, and overall project architecture.

After the planning phase, my primary responsibility was to design and implement frontend functionalities using Next.js. As a frontend developer, I focused on building responsive and interactive user interfaces, translating design mockups into code, and ensuring seamless integration with backend APIs. Frontend development included handling client-side logic, state management, and implementing features that directly affect the user experience, such as browsing trekking packages, booking interfaces, and navigation components.

The *Everest Adventure Nepal* web application was developed to simplify the process of booking Everest Base Camp trekking packages, including luxury and semi-luxury options. The application was designed to be user-friendly, responsive, and performant, enabling users to access trekking information, browse packages, and complete bookings from any device. Interface elements were optimized for accessibility and usability to improve overall user satisfaction.

Some key features implemented on the frontend of the *Everest Adventure Nepal* web application include:

- Responsive user interface design for different screen sizes
- User authentication and registration interfaces
- Dynamic trekking package listings and details
- Interactive booking and reservation forms
- Integration with REST APIs for data retrieval and updates
- Client-side validation and error handling

4.3.1 Development Methodology

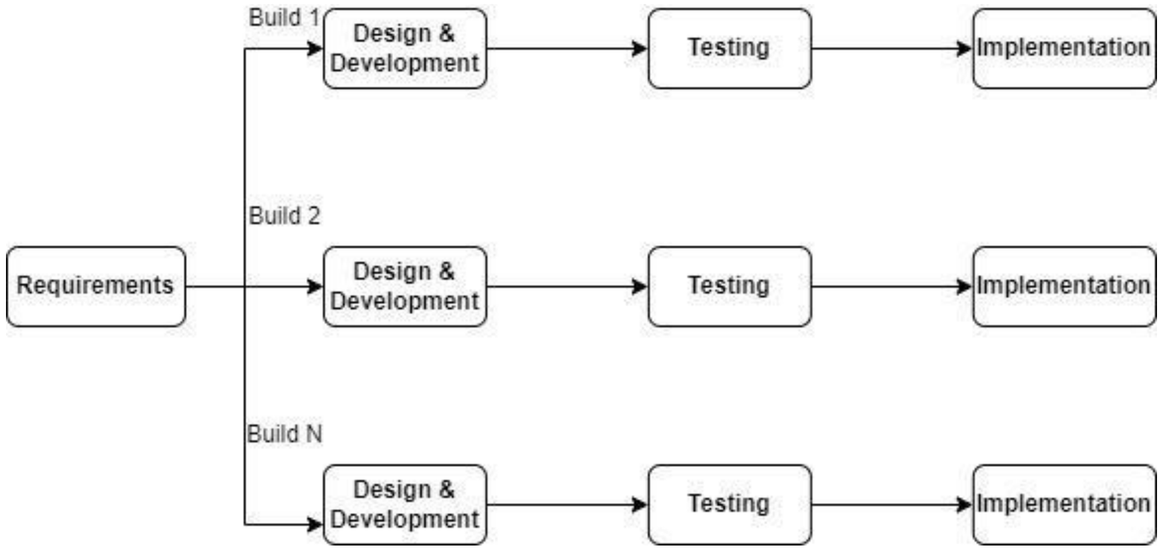

Figure 3: Incremental Model

An **Incremental Software Development Life Cycle (SDLC) model** was used for the development of this project. In this model, the system was developed in multiple increments, where requirements were analysed, designed, implemented, and tested in each phase. This approach allowed continuous improvement and early identification of issues during development.

1. **Requirement Analysis:**
   In this phase, existing systems were studied to understand their functionalities and limitations. The requirements necessary for developing the new system were identified and documented. Information was collected through various sources such as journals, research papers, articles, and existing applications. All relevant findings were carefully analyzed and recorded to define clear system requirements.

2. **Design and Development:**
   During this phase, the system architecture and design were prepared based on the gathered requirements. The requirements identified in the analysis phase were transformed into a detailed system design. Following this, development of the system prototype was carried out incrementally, focusing on implementing core functionalities in each iteration.

3. **Testing:**
   Each module was tested individually before being integrated into the overall system. Testing was performed to identify and fix coding errors, functional issues, and system inconsistencies. API integration and system compatibility were also verified to ensure

proper communication between different system components.

4. **Implementation:**

   After successful testing, the system was deployed and made available for use by the intended users. This phase ensured that the application functioned as expected in a real-world environment and met the defined requirements.

4.3.2 System Design

The system design phase of the Everest Adventure Nepal project was a critical stage that laid the foundation for successful frontend implementation. This phase involved making key decisions related to the frontend architecture, user interface structure, data flow, component organization, and integration with backend services. The primary objective was to design a scalable, maintainable, and user-centric frontend system that aligns with the project requirements and supports future enhancements. Frontend system design focuses on planning how the UI components, routes, state management, and AnPI communication work together to provide smooth and efficient user interaction.

The Everest Adventure Nepal system consists of the following main components:

- **User Interface (UI):**

  The User Interface provides an intuitive and user-friendly platform for users to interact with the application. It enables users to navigate the platform easily, browse trekking packages, view detailed itineraries, and complete bookings in a responsive and interactive environment. UI design decisions included layout structure, responsive behavior, component reuse, and visual consistency using technologies like Next.js, Tailwind CSS, and Shadcn UI.

- **Frontend Architecture (Next.js):**

  The frontend architecture is structured around reusable components, state management, and efficient data fetching from backend APIs. Next.js was chosen for its ability to deliver performance optimizations such as server-side rendering and static generation, improving initial load times and user experience.

- **API Integration:**

  The application frontend communicates with backend RESTful services to access data such as user profiles, trekking packages, booking information, and payment status. Proper API integration was designed to handle asynchronous data fetching, loading states, and error responses effectively.

- **Responsive & Adaptive Design:**

The system design ensured that the application is responsive and accessible across different devices and screen sizes. Tailwind CSS was used for utility-first styling to create flexible and consistent UI elements, while component libraries like Shadcn UI standardized the visual language across the application.

4.3.3 Tools and Technologies Used

The following software tools and programming languages were used in the development of the Everest Adventure Nepal project:

- **Git:**
  Git was used for version control to manage source code, track changes, and support collaborative development within the team.
- **Visual Studio Code (VS Code):**
  Visual Studio Code was used as the primary Integrated Development Environment (IDE) for writing, editing, and debugging the application code.
- **Next.js:**
  Next.js was used as the frontend framework to build a fast, scalable, and server-rendered web application.
- **Tailwind CSS:**
  Tailwind CSS was utilized for styling the user interface, enabling rapid development of responsive and modern designs using utility-first classes.
- **Shadcn UI:**
  Shadcn UI was used to implement reusable and customizable UI components, improving consistency and efficiency in frontend development.
- **Figma:**
  Figma was used for UI/UX design, prototyping, and visualizing the layout of the application before development.

Frontend

Next.js

Next.js is an open-source web development framework built on top of React that simplifies building modern, fast, and scalable user interfaces. It provides features such as file-based routing, server-side rendering (SSR), and static site generation (SSG) out of the box, which help improve performance, search engine optimization, and user experience compared to plain React applications. Next.js handles routing, rendering strategies, and performance optimizations automatically, enabling developers to focus on building responsive and interactive frontend features. In this project, Next.js was used to develop a responsive and efficient frontend that communicates with backend services through RESTful APIs.

Backend

Java

Java is a robust, object-oriented programming language widely used for building secure and scalable backend applications. In this project, Java with the Spring Boot framework was used to develop server-side logic, handle business processes, manage database operations, and create RESTful APIs for smooth communication with the frontend.

The backend of the project was developed using Java, Spring Boot, PostgreSQL, and REST framework, ensuring security, scalability, and efficient data handling.

Software Tools

- Visual Studio Code (VS Code): Used as the primary code editor for writing, debugging, and managing frontend code. VS Code is a lightweight, versatile text editor that supports JavaScript, HTML, CSS, and Next.js development.
- GitHub: Used for version control and code sharing among team members.

Documentation Tools

- **MS Word:** Used for preparing project documentation and internship reports.

4.4 Tasks/Activities Performed

At the beginning of the internship, I was assigned to learn the fundamentals of Next.js frontend development and modern web technologies, including JavaScript (ES6+), responsive design, component-based architecture, and RESTful API integration. I also learned about UI/UX implementation, state management, client-side routing, and frontend best practices using Next.js, Tailwind CSS, and Shadcn UI. During the internship, I worked on the *Everest Adventure Nepal* project as a Frontend Developer (Intern). My main responsibilities included developing and maintaining user interfaces using Next.js, implementing responsive and interactive UI components, integrating frontend components with backend RESTful APIs, handling client-side logic and validation, and ensuring smooth user experience across devices. I also collaborated with the design team to translate Figma prototypes into functional frontend features & tested frontend .

CHAPTER 5: CONCLUSION AND LEARNING OUTCOME

5.1 Conclusion

The *Everest Adventure Nepal* web application was developed to simplify and improve the process of booking Everest Base Camp trekking packages through a reliable and user-friendly digital platform. The system enables users to access detailed trekking information, view transparent pricing, and complete bookings conveniently from anywhere at any time.
This internship at Code Bridge provided me with valuable practical experience in frontend development using Next.js and modern web technologies. I gained hands-on exposure to UI/UX implementation, responsive interface design, API integration, and real-world development workflows. Throughout the internship, I collaborated with experienced developers, which enhanced my understanding of professional software practices, teamwork, and communication. During the internship, I developed technical skills in frontend frameworks and tools, improved my problem-solving abilities, and learned the importance of writing clean, maintainable code. This experience significantly strengthened my foundation in web application development and prepared me for future challenges in the IT industry. Overall, the internship played a crucial role in my professional growth, and the knowledge and experience gained will be highly beneficial for my future career in frontend development and IT.

5.2 Learning Outcome

Learning outcomes encompass the knowledge, skills, attitudes, and habits of mind acquired from a learning experience. Working at Code Bridge provided me with invaluable insights, helping me not only in career decision-making but also in recognizing my strengths and weaknesses. I am aware that the knowledge and experience gained during my time at Code Bridge will be immensely valuable as I embark on my career path shortly. Some outcomes are: -

- Working with multiple features under pressure to meet deadlines.
- Familiarize oneself with the professional working environment.
- Understand the organization's culture.
- Gained experience of how to work collaboratively within the group with a proper team.
- Time management and communication skills.

REFERENCES

Bass, L., Clements, P., & Kazman, R. (2013). *Software Architecture in Practice* (3rd ed.). Addison-Wesley.https://www.pearson.com/en-us/subject-catalog/

Gretzel, U., Sigala, M., Xiang, Z., & Koo, C. (2015). *Smart Tourism: Foundations and Developments*.https://link.springer.com/book/

Kaur, A., & Singh, K. (2020). Security Issues in Web Applications. *International Journal of Computer Applications*.https://www.ijcaonline.org/

Trekking Agencies' Association of Nepal (TAAN). (2023). Official Website. https://taan.org.np/

Bookmundi. (2023). Nepal Trekking Packages.https://www.bookmundi.com/

Viator. (2023). Tours and Activities in Nepal.https://www.viator.com/

GetYourGuide. (2023). Nepal Travel Experiences. https://www.getyourguide.com/

APPENDIX

Table 4: Supervisor's Visit Log Sheet

S no.	Date	Task Done	Remarks
1.	29/5/2082	Intern proposal defense	
2.	15/07/2082	Reviewed some internship tasks	
4.	5/8/2082	First internal defense	
5.	6/8/2082	Mid defense	
6.	5/10/2082	Reviewed the overall documentation	
7.	30/12/2082	Finalized the report	

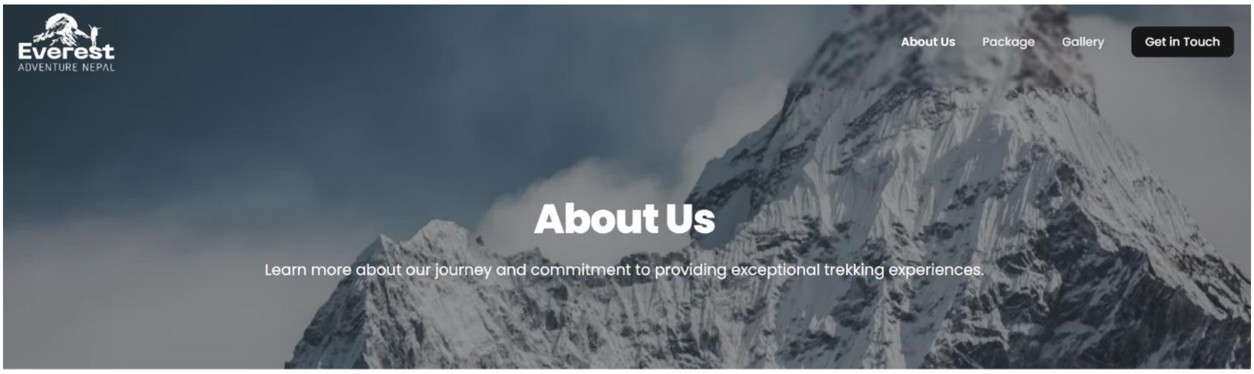

Figure of About us Frontend

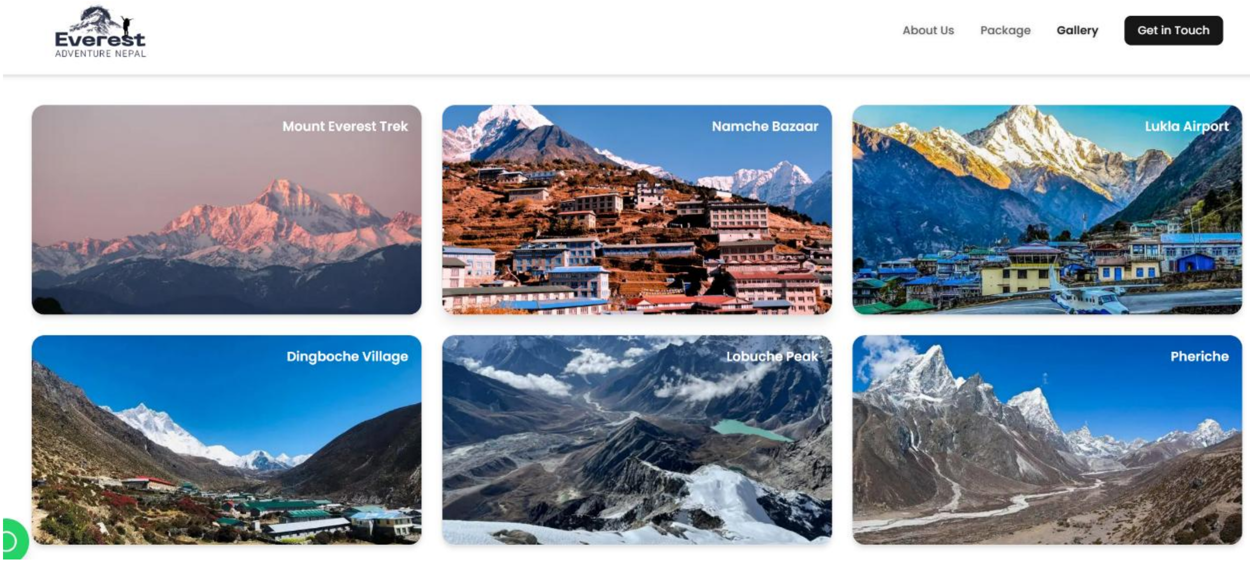

Figure of Gallery part

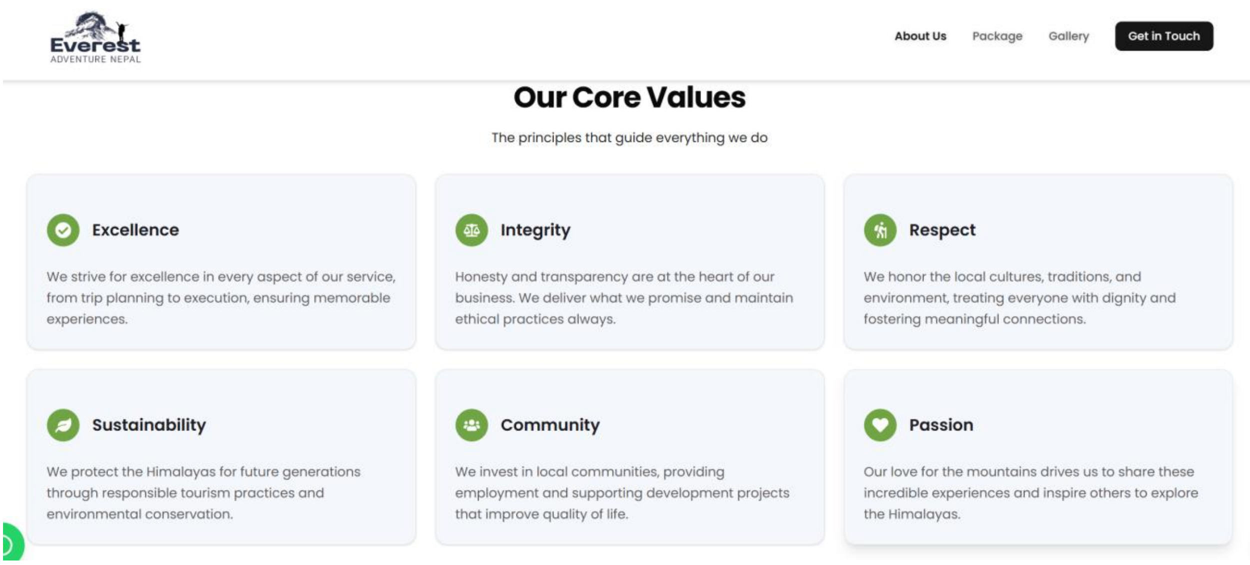

Figure of About core values

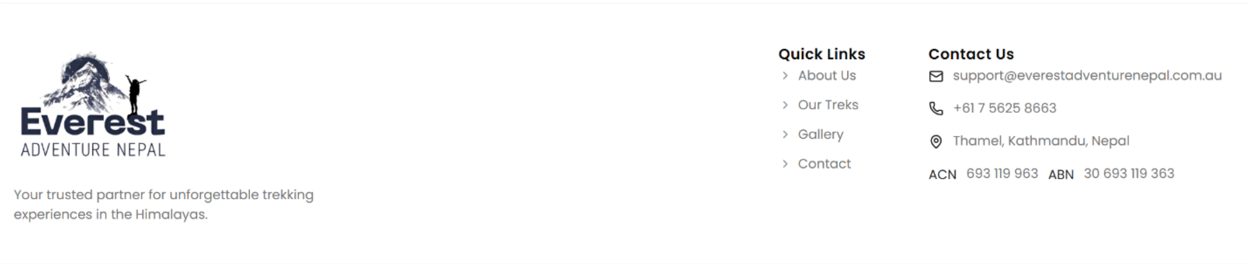

Figure of contact us

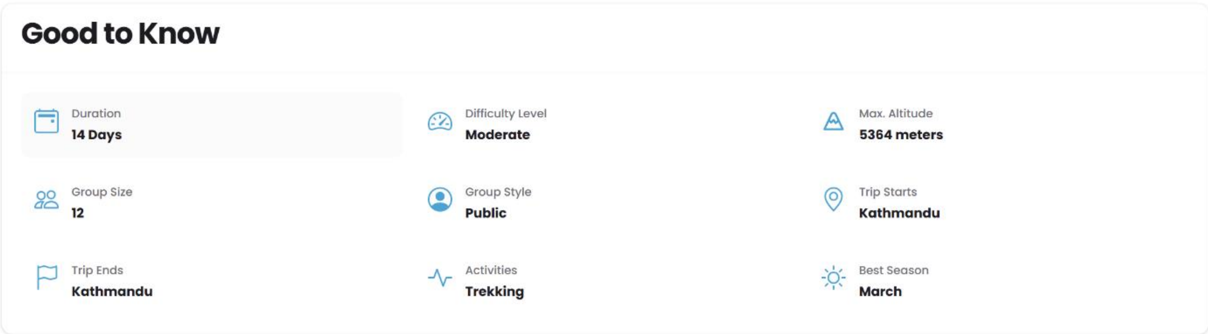

Figure of key highlights part

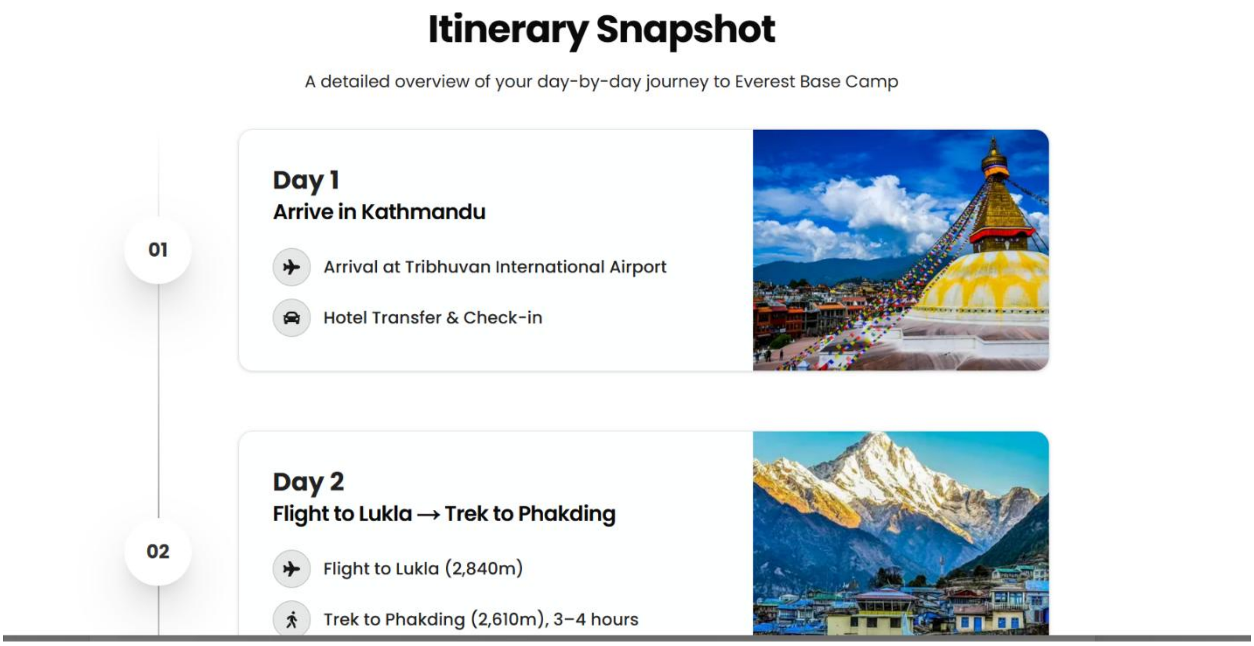

Figure of Itinery snapshot

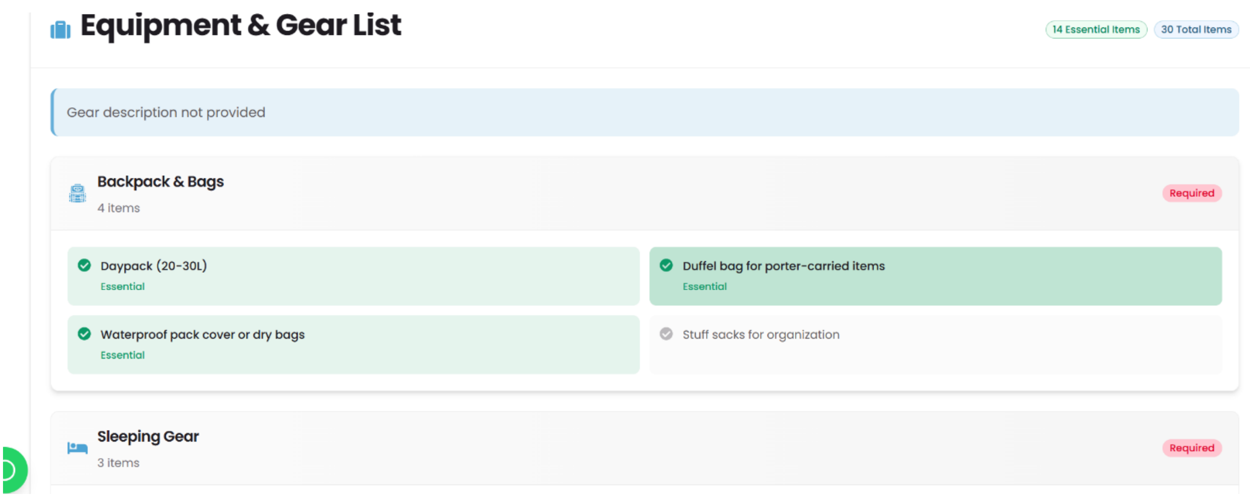

Figure of Customer Gearlist ideas